

Research

2D-QSAR driven design, molecular docking, molecular dynamics simulation and MM/GBSA studies on quinazoline derivatives for development of VEGFR-2 inhibitors

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Abstract

Various in silico approaches were utilized to design quinazoline derivatives as anticancer agents targeting VEGFR-2. Six 2D-QSAR models were generated via Monte Carlo optimization method of CORALSEA software. The models generated were based on hybrid optimal descriptors including Graphs and SMILES. All the models were observed to be robust via internal and external validation parameters. Model M4 of split 3 had the highest average correlation coefficient ($r^2=0.7209$), average R_m^2 value (0.6256) and lowest error (MAE=0.465). Structural features characterised as promoters and hinderers were extracted from the M4 model and their mechanistic interpretation was employed to improve the pIC_{50} of least potent compound SM33 from the dataset. Removal of hinderers and addition of promoters resulted in a series of compounds, DDSL6a-DDSL6l among which DDSL6j was the most active compound with a pIC_{50} of 8.07. All the compounds were subsequently subjected to molecular docking against PDB: 4ASD, with DDSL6j having better LF rank score (− 14.032) and LF dG score (− 12.318). MD simulation of DDSL6j, sorafenib and SM33 carried out at 200 ns revealed sustained interactions of these compounds with all the key amino acid residues of the protein. Improved potency of compound DDSL6j was attributed to presence of additional stable H-Bond interaction. MM/GBSA calculation also indicated stable complex with binding free energy of 0.28 kcal/mol. All the designed molecules were also found to be possess good ADMET profile. These findings indicate that the quinazoline nucleus can be investigated and refined to create novel VEGFR-2 inhibitors as potential anti-angiogenic drugs in future.

Keywords Quinazoline · QSAR · Monte Carlo · VEGFR-2 · Molecular docking · MD simulation

1 Introduction

Vascular endothelial growth factor receptor-2 (VEGFR-2) is a tyrosine kinase receptor present on endothelial cells, and is responsible for angiogenesis [1]. Angiogenesis plays a major role in physiological and pathological functions like wound healing, inflammation, development of foetus and transformation of benign tumor into malignant. It is exaggerated by various signals including hypoxia, cellular injury and secretion of angiogenic stimulators like VEGF by cancer cells [2]. Some cancers particularly colorectal and renal cancers are characterized by high levels of VEGF. Hence, VEGFR-2 is

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considered as a major target for treatment of cancer [3]. Eleven small molecule VEGFR-2 inhibitors have been approved by FDA (Food and drug administration) among which vandetanib is the first quinazoline-based molecule. It is developed in 2011, and is recommended for treatment of late-stage medullary thyroid cancer [4]. Recently, fruquintinib has been approved in China for the treatment of metastatic colorectal cancer [5]. Cediranib is another drug, which is currently under phase 3 clinical trial [6]. Despite being potent, these drugs pose some side effects like hypertension, CVS-related diseases, hepatotoxicity and drug resistance (Table 1). Hence, there is a need for novel small molecule VEGFR-2 inhibitors that can overcome these limitations.

Quinazoline belongs to the class of nitrogen-containing fused aromatic heterocyclic nuclei, and chemically known as benzo[e]pyrimidine. Since the discovery of first quinazoline-based natural alkaloid (peganine), which is a bronchodilator, researchers have explored the biological potential of this nucleus in various therapeutic fields including antimalarial, antiprotozoal, antihypertensive, GABAergic and most essentially anticancer in the form of tyrosine kinase inhibitors (Fig. 1) [8, 9].

Moradi et al. reported a detailed review on quinazoline derivatives as VEGFR-2 inhibitors focusing on its SAR analysis to explore the structural features required in designing potent inhibitors with good potency and minimum side effects [10]. Recently, reports on several benzo-fused heteronuclear compounds as VEGFR-2 inhibitors available in literature have also been critically reviewed [11]. These reports reveal that occupancy of quinazoline moiety in the ATP-binding site is responsible for inhibition of the receptor. Some of the potent quinazoline-based VEGFR-2 inhibitors reported in literature are shown in Fig. 2. Compound 1 has the inhibitory activity (IC_{50} 2.5 μ M) similar to that of sorafenib (IC_{50} 2.4 μ M) [12], whereas compound 2 exhibits activity better than sorafenib [13]. In 2022, Hamdi et al. developed substituted 4-anilino-2-vinylquinazolines, and compared their cytotoxic activity. Among those, compound 3 displays an activity comparable to sorafenib [14]. Another research group has investigated the inhibitory potential of quinazoline-derived molecules against

Table 1 Quinazoline based FDA approved VEGFR-2 inhibitors

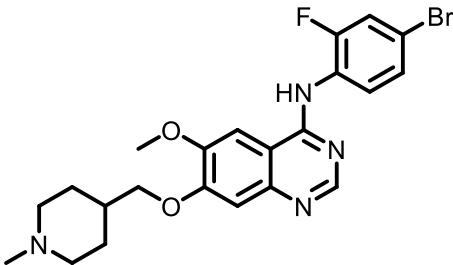
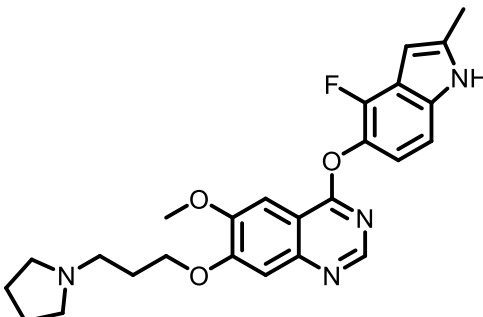
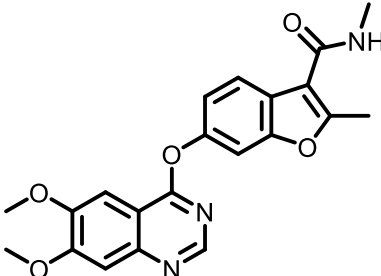
Name	Structure	Uses	Side effects
Vandetanib		Medullary thyroid cancer (MTC)	• Hypertension • Heart failure • Arrhythmia • Blurred vision
Cediranib		Ovarian cancer	• Anorexia • Skin rash • Fatigue • Diarrhoea
Fruquintinib		Metastatic colorectal cancer	• Bleeding gums • Blurred vision • Cloudy urine • Depression

Fig. 1 Biological activity profile of quinazoline-based compounds

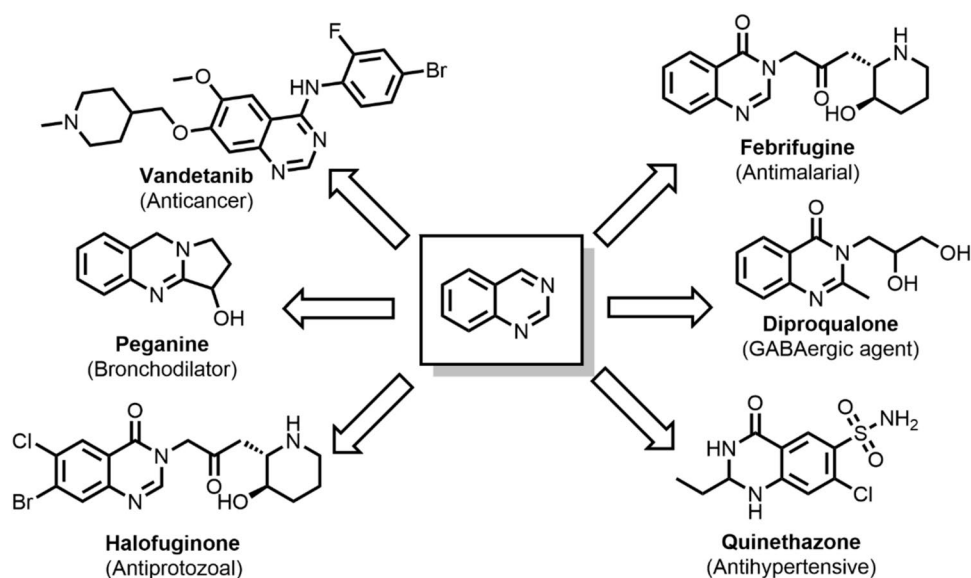
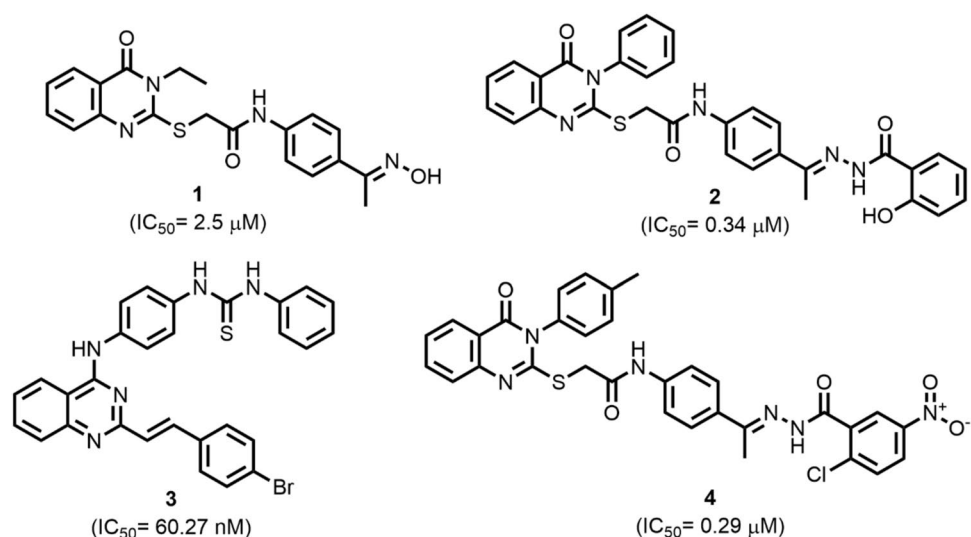


Fig. 2 Some potent quinazoline based VEGFR-2 inhibitors reported in literature



VEGFR-2, and disclosed that derivatives containing electron-withdrawing groups like chloro and nitro, as exemplified by compound **4**, show significantly better activity than sorafenib (Fig. 2) [15].

Traditional drug development process is an analog design type of approach wherein a medicinal chemist designs molecule based on the existing structure–activity relationship data and its own intuition, synthesizes it, and then evaluates for the activity using appropriate in vitro and in vivo models. This process takes around 10–15 years of time and costs more than 1 billion USD to come up with a molecule that can be registered as a drug. The major limitations of this approach include hit and trial method, excessive usage of chemicals and animals, and high chances of failure at later stages due to toxicity and/or side effects. In silico approaches in drug design have emerged as indispensable tools not only to speed up the drug design and development process but also to lower the cost of drug development as well as fewer chances of failure at later stage [16, 17]. Computer aided drug design (CADD) is a set of computational techniques that assists researchers in drug designing by analysing various structural features of protein, genetic material, enzymes, drug-receptor interaction, etc. [18]. Docking analysis is one of the techniques which is primarily done for screening out the designed compounds based on their affinity with the target followed by molecular dynamics (MD) simulation which helps in assessing the stability of the drug molecule within the active site of the target. Quantitative structure activity relationship (QSAR) is another in silico approach for designing compounds. It utilizes different algorithms based on the software to quantify the relationship between structural features of a chemical compound and its biological

activity. Based on the extracted features required for optimum activity, new compounds are designed and their activity is predicted. Various QSAR studies have been performed on quinazoline nucleus as anticancer agent targeting EGFR, thymidylate synthase, TLR7, histone deacetylase 6 and PI3K α but no such study has been conducted on VEGFR-2 as of today [19–23].

The current study was designed to develop 2D-QSAR models using CORALSEA software based on Monte Carlo algorithm for identification of structural features essential for VEGFR-2 inhibition. On the basis of features extracted, 17 new quinazoline derivatives were designed and their VEGFR-2 inhibition activity was predicted. Binding interaction of the designed compounds was studied using Cresset, and then on the basis of docking score and predicted activity, the best compound was studied for MD simulation. Moreover, the ADMET properties of the designed compounds was predicted via Deep-PK server.

2 Materials and methods

2.1 Dataset

Literature survey was performed to collect 288 quinazoline derivatives acting as VEGFR-2 inhibitors [24–44]. The structures were drawn in ChemDraw professional and converted into canonical SMILES by using OpenBabel GUI program [45]. Compound ID SM224 was found to be duplicate SMILES so it was removed from the dataset. In vitro VEGFR-2 inhibitory activity (IC_{50}) of these molecules was used as dependent variable which vary across the nM to μ M range. To standardize the data, IC_{50} values were converted into pIC_{50} (i.e., $pIC_{50} = x - \log IC_{50}$) values.

Three random splits were generated, each divided into four subsets:

1. Active training set (+): Molecules from this set were used for generating the model by calculating the correlation between the molecular descriptors and their end point.
2. Invisible training set (–): Prevents overfitting by keeping additional data aside.
3. Calibration set (#): To evaluate the performance of the model.
4. Validation set (*): To assess the reliability and predictive potential of the model generated.

The percentage identity of each split and distribution of compounds in different subsets of each split is shown in Table S1 and Table S2 (Supplementary data) respectively.

The percentage identity was calculated using the Eq. 1 [46],

$$Identity(\%) = \left[\frac{N_{pq}}{(N_p + N_q)0.5} \right] \times 100 \quad (1)$$

N_{pq} number of compounds that are divided into the same subset for both p and q splits N_p number of compounds that are divided into the subset of p split N_q number of compounds that are divided into the subset of q split

2.2 Optimal descriptor

The CORALSEA software is based on using simplified molecular input line entry system (SMILES) and molecular graphs for determining the molecular descriptors. Hybrid of these two optimal descriptors was used in present study which increases the accuracy and validity of the model generated.

The DCW of SMILES and molecular graph-based descriptors was calculated by using the following equations [47, 48]. The in-depth information of these descriptors is explained in previous study [49].

$$DCW^{SMILES}(T, N_{epoch}) = \sum CW(BOND) + \sum CW(NOSP) + \sum CW(HALO) + \sum CW(HARD) + \sum CW(ATOMPAIR) + \sum CW(Sk) + \sum CW(SSk) + \sum CW(SSSk) \quad (2)$$

$$DCW^{Graph}(T, N_{epoch}) = \sum CW(Ak) + \sum CW(^0EC_k) \quad (3)$$

The combined DCW of these two molecular descriptors was calculated by following the equation,

$$DCW^{Hybrid}(T, N_{epoch}) = DCW^{Graph}(T, N_{epoch}) + DCW^{SMILES}(T, N_{epoch}) \quad (4)$$

2.3 Model development

The CORALSEA software works on Monte Carlo algorithm which relies on threshold (T) and number of epochs (N_{epoch}), the two basic parameters. The T value classifies the structural features into active and rare attributes (SAk). If the frequency of an attribute in the active training set (+) is less than the T value, then it is considered as rare feature and not considered in the modelling process as they would just act as noise.

By following the previously studied methodology [49], the recommended T and N_{epoch} value was calculated. All the models were generated in triplicate and best model was selected based on the average value of all the important parameters like R^2 , Q^2 etc.

The VEGFR-2 inhibition was calculated by using the equation [50],

$$pIC_{50} = C_0 + C_1 \times DCW^{Hybrid}(T^*, N^*) \quad (5)$$

C_0 and C_1 are the regression coefficients, while T^* and N^* are the recommended values of threshold and epoch respectively. These values were determined based on maximum value of r^2 achieved by calibration set.

2.4 Validation of QSAR model

Validation of the developed models is a very important criteria in QSAR analysis to evaluate its reliability, robustness and predictive potential. Three types of validation can be performed internal validation, external validation and Y-randomisation. In internal validation, Leave-One-Out (LOO) technique was used as detailed out in the previous study [49]. As mentioned earlier, the LOO cross-validation parameter is not sufficient to assess the predictive ability of QSAR models, and therefore application of external validation is essential. External validation determines the novel metric r_m^2 that was used to validate true predictive potential of QSAR models [51]. Y-randomization test was also performed to confirm the robustness of the developed QSAR model, ensuring that it is not due to chance. In this test cR_p^2 should be > 0.5 for robust model [52].

2.5 Applicability domain (AD)

Applicability domain helps in assessing the reliability of the developed model. It is constructed by utilizing the compounds in the training set. It defines a biological, structural or physico-chemical range within which the QSAR model can make predictions for compounds that were not included in the dataset used for generating the model. Compounds which are not within the AD range are considered as outliers i.e., their prediction will not be accurate. This may be due to their structural variation from compounds which fall in AD range [53]. A compound is considered as an outlier if it fall outside the applicability domain (AD) limits, i.e. if:

$$DefectSMILES < 2 \times \overline{DefectSmiles} \quad (6)$$

where $DefectSMILES = \sum SAK - Defect$

2.6 Molecular docking

The binding interaction of the newly designed compounds was studied using Flare module of Cresset software against PDB: 4ASD which was obtained from protein data bank and evaluated on the basis of the software's Lead Finder (LF) algorithm. Prepare-protein wizards tool of the Flare module was for protein preparation which involves basic steps like modeling of missing loops, addition of missing atoms, protonation of titratable residues by applying XED force field at pH of 7.4 and removal of co-crystallized ligand and the exterior water. The same force field was used to prepare the designed ligands and minimize to generate their best conformation. The A BAX 1500_D (Sorafenib) is the co-crystallized ligand which defines the binding site and generate an energy grid map of 6 Å radius. Sorafenib was redocked into the protein to confirm the reliability of docking algorithm utilised in the study. All the designed compounds were docked and were scored using LF scoring functions with normal calculation method. The LF algorithm calculates three types of scores: rank score, dG score and VSscore. The rank score indicates how much the pose is closer to the crystallographic structure. dG score evaluates binding affinity and VSscore indicates how much is the ligand found to be active. More the negative these values better are their respective characteristics. The compound to be selected for further analysis via MD simulation was based on the three LF scores generated by the Flare software and pIC_{50} calculated from the best fit model [54].

2.7 Molecular dynamics (MD) simulation

The stability and behaviour of the best docked compound within the VEGFR-2 active site was analysed via means of MD simulation using AMBER GAFF2 (Flare 8.0, Cresset package). The protein–ligand combination served as the centre of the solvent system, which was built around it using the explicit solvent water model. The box's dimensions ($10 \times 10 \times 10$ Å) were kept in truncated octahedron shape. The protein–ligand complex was then reduced to a minimum by equilibrating each complex for 200 ps and establishing the minimum energy tolerance at 0.25 kcal/mol. 200 ns was used as the recording time and a time step of 4.00 fs was used for the simulation. Using the Root Mean Square Deviation (RMSD) plot and Root Mean Square Fluctuation (RMSF) plot, the complex stability was examined and contrasted before and after the simulation [54].

Additionally, Dynamic Cross-Correlation Matrix (DCCM) of the C- α chain displacement was performed to assess the correlated and anti-correlated movements of the protein residues before and after MD simulation [55]. The DCCM plot was computed, analysed, and visualized using Python libraries in Jupyter Notebook by following code provided in the supplementary information. For trajectory processing and structural analysis, MDAnalysis library was utilized [56]. The analysis focused on C- α atoms of residues 807–1168 from both pre- and post-simulated structures. The Pearson correlation coefficient was employed to compute the DCCM based on standard protocol [57]. NumPy was used for matrix calculations, while Matplotlib and Seaborn were utilized to generate heatmap visualizations.

2.8 MM/GBSA

The calculation of the free energy of ligand–protein binding of most active compound with VEGFR-2 was performed using Molecular Mechanics, General Born Surface Area (MM/GBSA) of flare software by using the equation given below. Single trajectory protocol was used which is much faster than separate trajectory protocol. The solvent model was kept as Implicit GBn2 and both the protein and ligand were minimized [58].

$$G_{bind} = E_{bond} + E_{elec} + E_{vdW} + G_{pol} + G_{non-pol} - TS \quad (7)$$

Here, G_{bind} = binding free energy of receptor and ligand, E_{bond} = bond energy, E_{elec} = electrostatic energy, E_{vdW} = van der Waals energy, G_{pol} = polar energy, $G_{non-pol}$ = difference between non-polar and polar energy, TS = conformational entropy.

2.9 ADMET profiling

The prediction of ADMET properties of drug molecules have become a crucial part in drug discovery as the molecules may possess good pharmacological activity but does not have the suitable properties to dispense into a formulation or possess side effects and toxicity. Deep-PK server was used in present study to predict the ADMET of designed compounds DDSL6(a-l). It is very commonly used webserver for calculating the various ADMET properties like oral bioavailability, intestinal absorption, carcinogenicity etc. [59].

3 Results and discussion

3.1 Model validation

Six QSAR models were generated and compared. The statistical values of various parameters (R^2 , CCC, IIC, CII, Q^2 , Q^{*2} , MAE) for these models is provided in Table S3 (Supplementary data). Although all the models satisfied these parameters in the acceptable range, M3(HFG combined with zero order Morgan extended connectivity index and Atom pair) and M4 (GAO combined with zero order Morgan extended connectivity index and Atom pair) had the highest values of R^2 , CCC, Q^2 , Q^{*2} and among the two, M4 was concluded as the best model as it had better statistical values. The graphical representation of each split of best fit model M4 is shown in Figure S1 (supplementary data). Y- randomisation test was also conducted which displayed $cR_p^2 > 0.5$ (Table S4, supplementary data). The statistical validation parameters for predictability criteria for the QSAR model M4 in each split is provided in Table 2. Split three of model M4 had the highest value for average R_m^2 value which is an important criterion to demonstrate reliability and consistency of the model. Further M4 was used to calculate the pIC_{50} of the newly designed compounds by using the Eqs. (8–10)

Run 1:

$$pIC_{50} = -1.9190573(\pm 0.0520769) + 0.1028018(\pm 0.0005974) * DCW(3,12) \quad (8)$$

Run 2:

$$pIC_{50} = -0.2302149(\pm 0.0476017) + 0.1203085(\pm 0.0007808) * DCW(3,13) \quad (9)$$

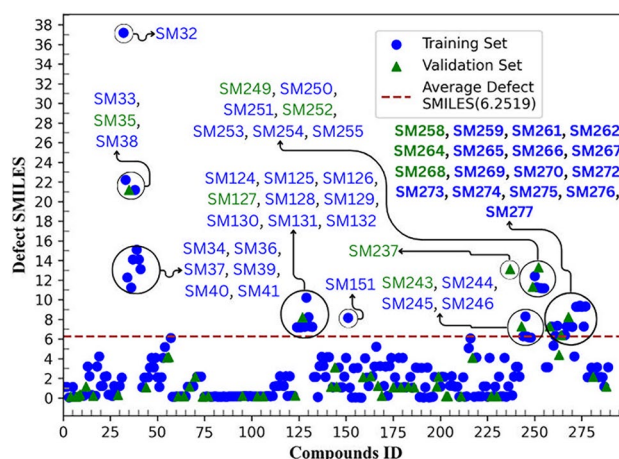
Table 2 Predictability criteria for the QSAR Model 4 for each split

Validation parameter	Split-1		Split-2		Split-3		Acceptance interval
	$r_v^2(x, y)$	$r_v^2(y, x)$	$r_v^2(x, y)$	$r_v^2(y, x)$	$r_v^2(x, y)$	$r_v^2(y, x)$	
r^2	0.6907	0.6907	0.6823	0.6823	0.7325	0.7325	> 0.5
r_0^2	0.6790	0.6378	0.6216	0.6716	0.6807	0.7284	> 0.5
rr_0^2	0.6378	0.6790	0.6716	0.6216	0.7284	0.6807	> 0.5
$\frac{(r^2 - r_0^2)}{r^2}$	0.0169	0.0766	0.0891	0.0158	0.0707	0.0056	< 0.1
$\frac{(r^2 - rr_0^2)}{r^2}$	0.0766	0.0169	0.0158	0.0892	0.0056	0.0707	< 0.1
k	0.9957	0.9975	1.0045	0.9901	0.9690	1.0256	0.85 < k < 1.15
kk	0.9975	0.9957	0.9901	1.0045	1.0256	0.9690	0.85 < kk < 1.15
$r_m^2(test)$	0.6160	0.5318	0.5141	0.6115	0.5658	0.6854	> 0.5
Avg. R_m^2	0.5739		0.5628		0.6256		> 0.5
ΔR_m^2	0.0842		0.0974		0.1196		< 0.2

$r_v^2(x, y)$ y – experimental values; x – calculated values

$r_v^2(y, x)$ x – experimental values; y – calculated values

Fig. 3 Graphical representation of Applicability domain



Run 3:

$$pIC_{50} = -0.6181800(\pm 0.0448668) + 0.1055377(\pm 0.0006127) * DCW(3,12) \quad (10)$$

3.2 Applicability domain

The result of the applicability domain of M4 for three independent runs along with average value is provided in Table S5 (Supplementary data).. The average Defect SMILES was found to be 6.2519. So, by using the Eq. (6), out

Fig. 4 Selected potent compounds from dataset showing the presence of promoters (Green colour)

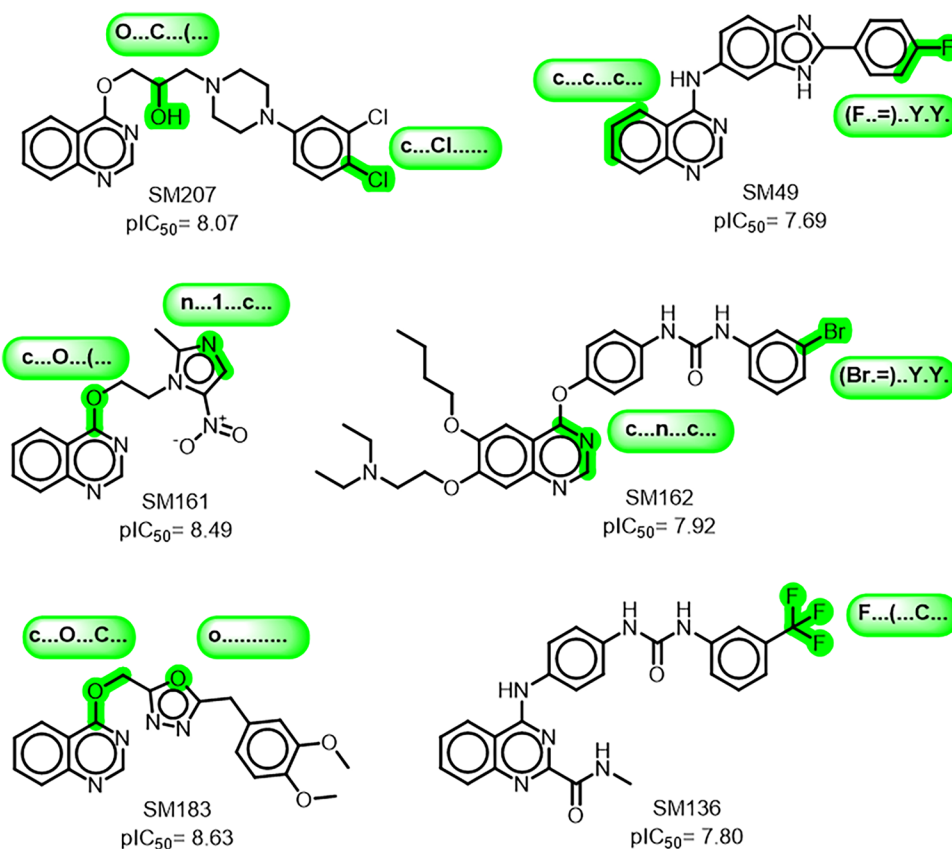
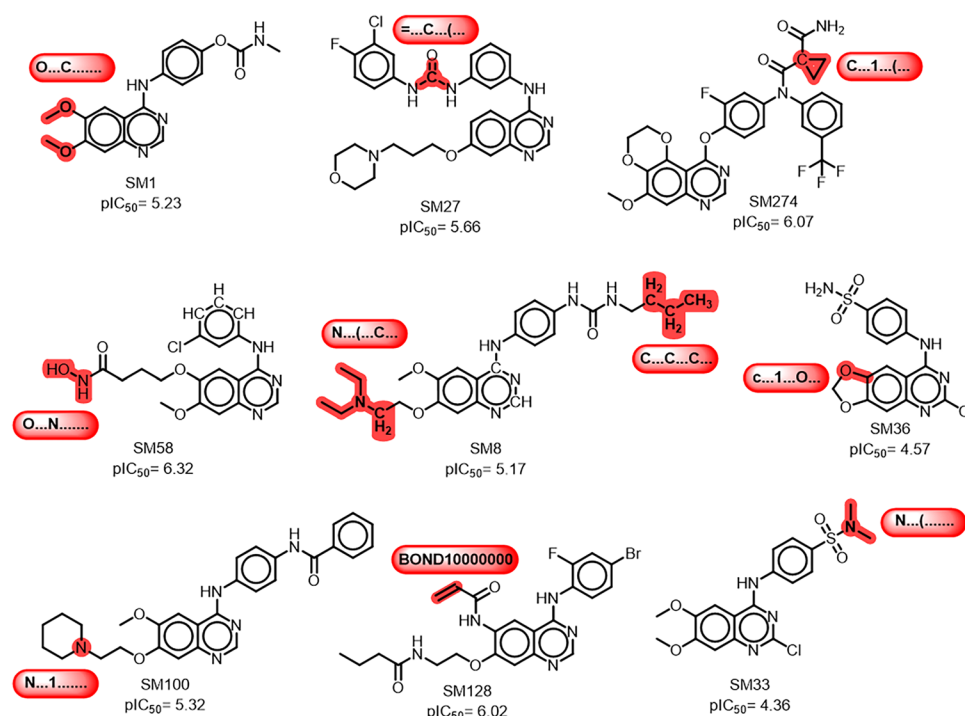


Fig. 5 Selected less potent compounds from dataset showing the presence of hinderers (Red colour)



of 288 compounds in total set only 49 came out to be outliers which concludes that predictive potential of M4 is 82.9%. This shows good reliability and robustness of the model. Figure 3 illustrates the graphical representation of the applicability domain and all the compounds in the dataset which fall in and out of the AD range.

3.3 Mechanistic interpretation

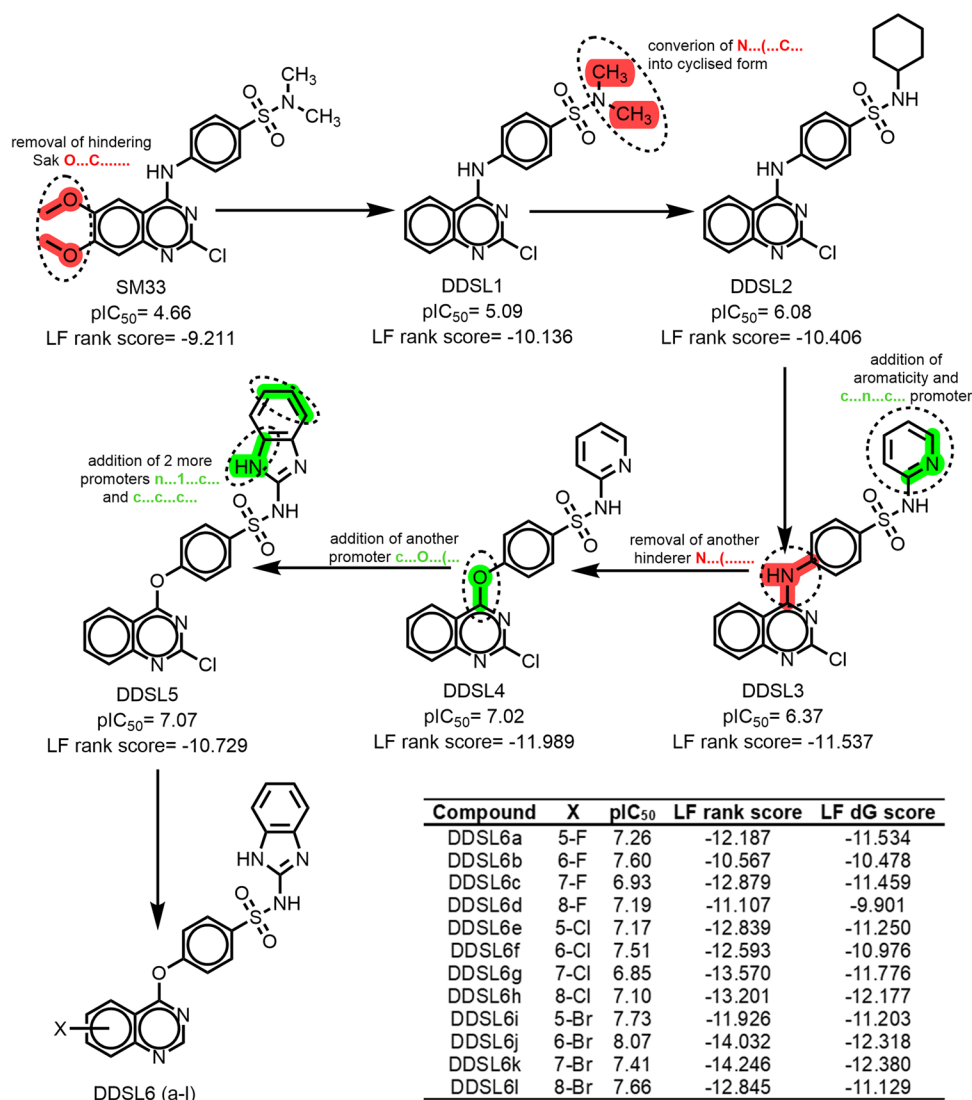
Various structural features were extracted from the M4 model by performing three independent runs. TableS6 (Supplementary data) represents various structural features labelled as promoters and hinderers. The features extracted from Monte Carlo optimization and pharmacophoric features mentioned in literature were then used to modify the least potent compound (SM33) from the dataset and designed a new series of compounds. Figures 4 and 5 demonstrates structural representation of various promoters and hinderers extracted from model M4 respectively.

As illustrated in Fig. 6, initially removal of O...C..... hinderer from SM33 resulted in increase in pIC₅₀ from 4.66 to 5.09. Removal of another hindering Sak N...(...), further enhanced the activity (pIC₅₀ 5.09–6.08). Introduction of aromatic ring (pyridine) on the nitrogen atom slightly increased the pIC₅₀ from 6.08–6.37. The replacement of non-aromatic nitrogen with oxygen also resulted in increase in pIC₅₀ (6.37–7.02). Final modifications including replacement of pyridine ring with a well reported pharmacophore (benzimidazole), removal of chlorine atom on quinazoline moiety and attachment of less electronegative atom bromine at 6th position of quinazoline ring resulted in significant increase in pIC₅₀ from 7.02–8.07. Further substitution with other halogens at different position of quinazoline ring was carried out to design the series DDSL6 (a-l). Among this series, presence of bromine at 6th position exhibited the highest potency. This elucidates that the promoters and hinderers that were extracted from the best fit model (M4) were valid.

3.4 Docking results

The binding affinity of the modified compounds at each step along with SM33 was carried out using Cresset flare software against PDB: 4ASD. Sorafenib was used as the reference compound as it is the co-crystallised ligand of the PDB used. Docking method was validated via redocking of Sorafenib into VEGFR-2 crystal structure. As revealed in Figure S2 (Supplementary data) conformation of redocked ligand possess identical orientation with the original co-crystallised ligand, with RMSD less than 2 Å⁰. This confirms the reliability and robustness of the performed docking analysis. Different types of scores i.e., rank score, dG score and VScore obtained for designed compounds are shown in Fig. 6. The 2D pose of all the compounds along with their docking score is shown in Table S7 (Supplementary data). The designed compounds

Fig. 6 Design of new compounds by structural modifications along with their corresponding pIC_{50} with different substitution pattern



exhibits majority of the necessary interactions with key amino acids which are essential for the affinity towards VEGFR-2. These residues include Asp1046, Phe1047, Leu889, Cys919, Cys1045 and Cys1024. Initially SM33 had one H-bond with Glu885 and some hydrophobic interactions which corresponds to its low LF rank score of -9.211. Removal of O...C... hinderer resulted in slight increase in LF rank score to -10.136 (DDSL1) which is due to formation of H-bond between Asp1046 and N-atom of quinazoline ring and some hydrophobic interactions like Val916, Val914, Lys868. Further modifications like removal of hinderers and addition of promoters significantly increased the LF rank score from -10.136 (DDSL2) to -14.032 (DDSL6j) which may be attributed to the addition of four H-bonds essential for VEGFR-2 inhibitory activity. The final designed compound i.e., DDSL6j had higher score than the reference drug, Sorafenib (-13.079). The 3D poses of the sorafenib, compound SM33, and DDSL6j with VEGFR-2 is shown in Fig. 7.

3.5 MD simulation results

Conformational stability of the most potent designed compound (DDSL6j), least potent compound selected from data set (SM33) and reference drug (Sorafenib) was evaluated within the active site of VEGFR-2 via conducting MD simulation study over 200 ns. The RMSD graph shown in Fig. 8 indicates that during the initial 25 ns the RMSD value increase rapidly to 3.5 Å as the system seeks an equilibrium state. After reaching an equilibrium state, there are minor stable fluctuations in RMSD up to 100 ns in the range of 3.0–3.5 Å which indicates the system is trying to attain a stable conformation. During the period of 100–200 ns the system has reached maximum stability. During the time period of 75–100 ns, the fluctuations

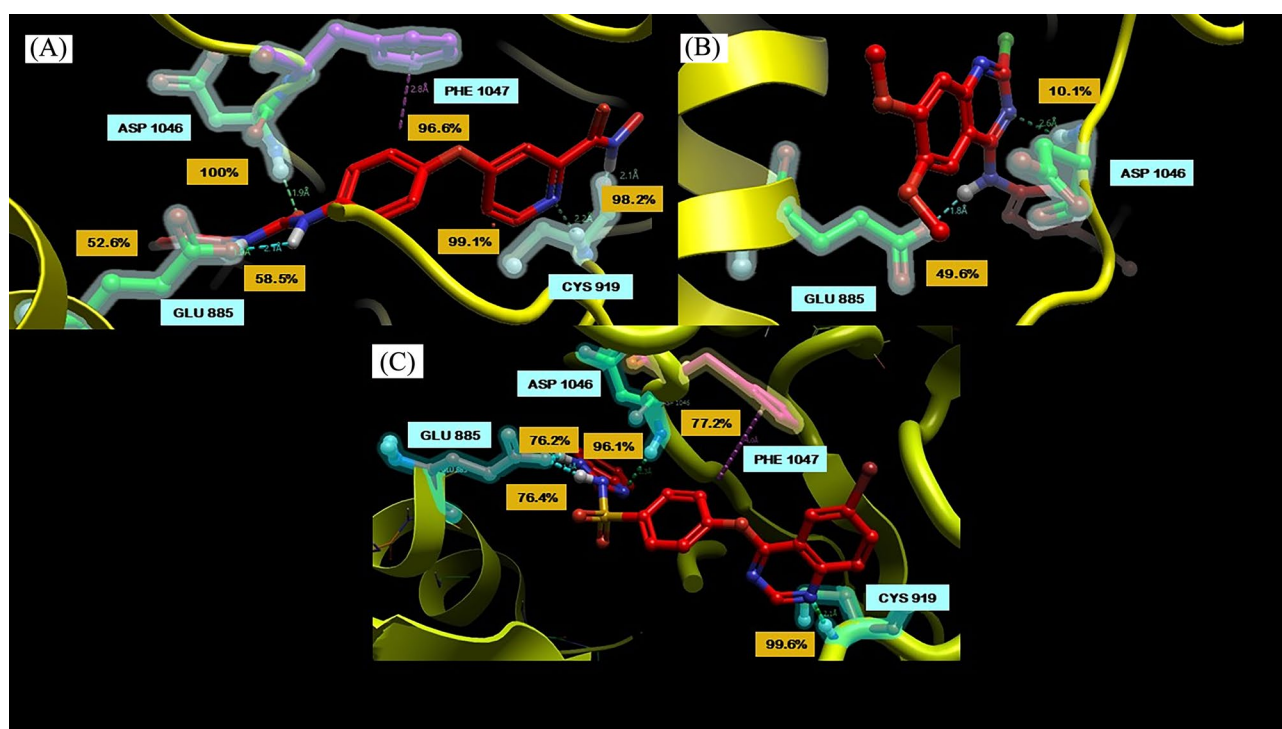
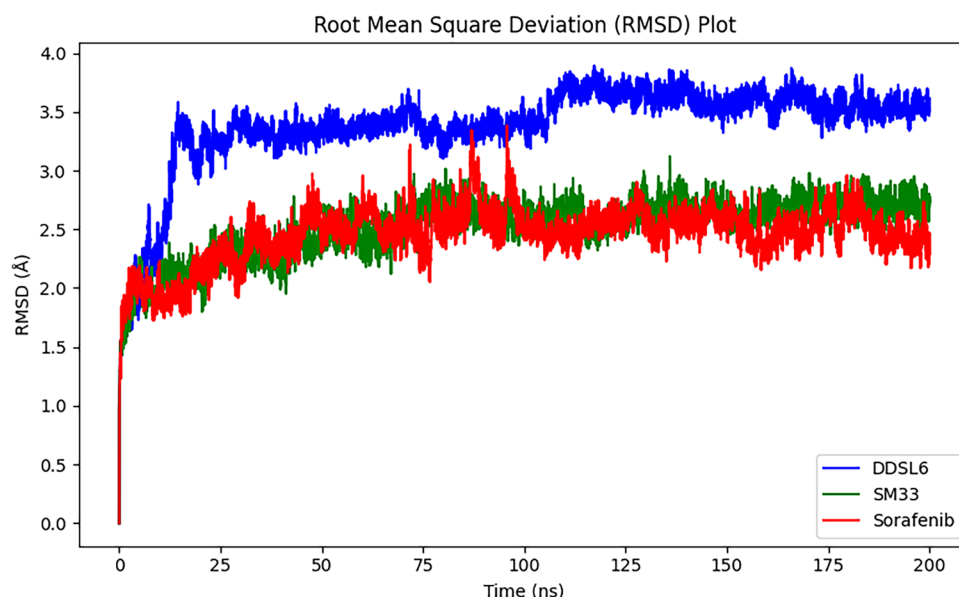


Fig. 7 3D poses of most potent compound DDSL6j (A), SM33 (B) and Sorafenib (C) in the active site of PDB: 4ASD (H-bond in green, π - π interaction in purple and % frame in yellow)

Fig. 8 RMSD plot of DDSL6j-receptor complex, SM33-receptor complex and Sorafenib-receptor complex for 200 ns MD simulation



are much more intense in case of sorafenib than DDSL6j. This indicates that the complex formed by the DDSL6j-receptor is more stable than the complex formed by sorafenib-receptor. Also, in the case of SM33 there is gradual increase in RMSD from 1.5 to 2.8 Å across the simulation period which suggests unstable interactions between the receptor and the molecule. Figure S3, S4 and S5 (Supplementary data) represents the % frames present of various amino acids interacting with the DDSL6j, SM33 and sorafenib in the VEGFR-2 active site respectively. In case of the DDSL6j, Cys919 and Asp1046 were found to be the most stable interactions with 99.6 and 96.1% frames which indicate very stable H-bond formation

Fig. 9 RMSF plot of DDSL6j-receptor complex, SM33-receptor complex and Sorafenib-receptor complex for 200 ns MD simulation

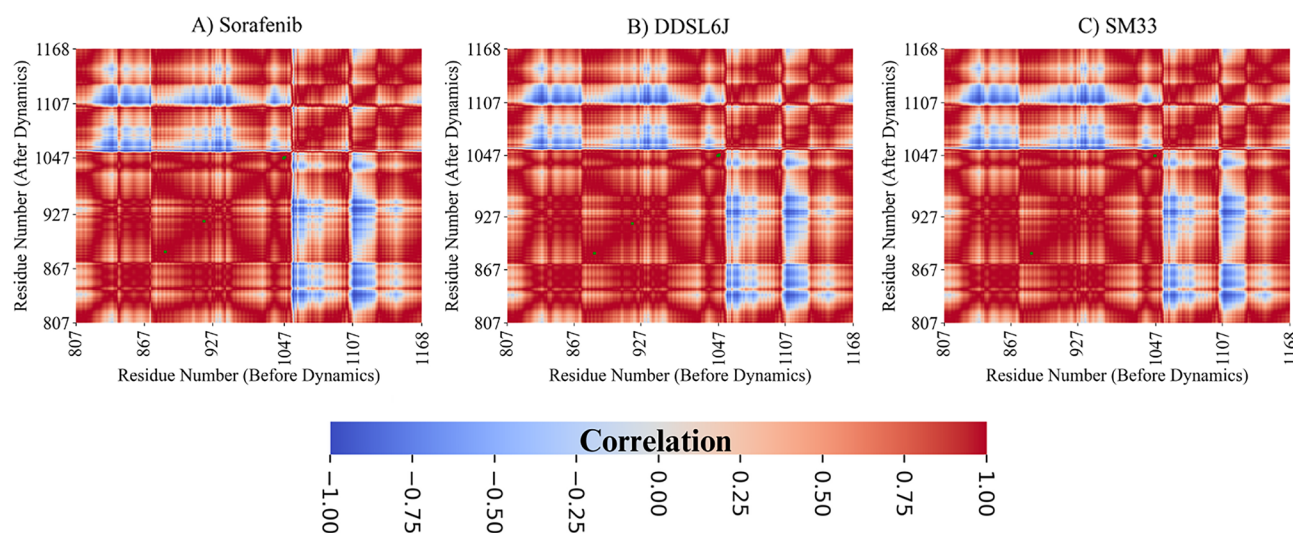
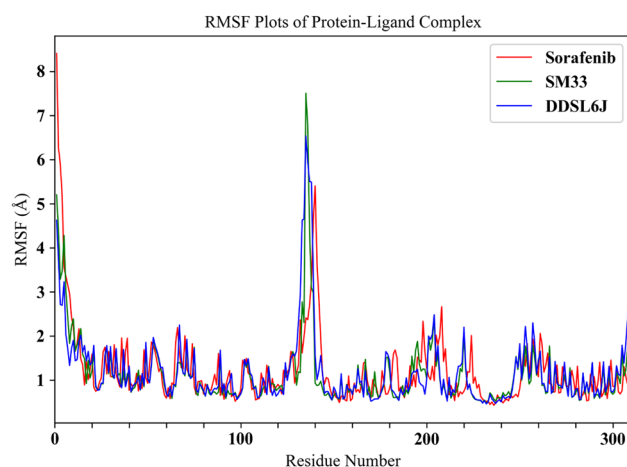


Fig. 10 The dynamic cross-correlation map of VEGFR-2 receptor (C α -atoms): **(A)** VEGFR-2/Sorafenib, **(B)** VEGFR-2/DDS6J, **(C)** VEGFR-2/SM33. Blue (– 1 to 0) and red (1 to 0) regions correspond to anti-correlated and correlated motion of the atoms, while white region suggests no correlation between the residues

Table 3 Energy components and their values in MM/GBSA calculations for DDSL6j-receptor complex

Total DG	BOND	ANGLE	DIHED	VD WAALS	EEL	1–4 VDW	1–4 EEL	EGB	ESURF
0.28	0.00	0.00	0.00	– 0.01	– 92.99	0.00	0.00	93.27	0.00

with N-atom of the quinazoline and benzimidazole moieties respectively. In addition, Glu885 also formed two H-bonds with similar level of percentage frames of 76.4 and 76.2%. Phe1047 had a percentage frame of 77.2% which suggests a strong and consistent π - π interaction between the aromatic ring of the ligand and protein. Among the key hydrophobic interactions, Val916 was observed to have the highest percentage frame of 87.6 and 84.9%. Where as in case of SM33, only Glu885 was found to be H-bonded with the molecule, that too even at a very low % frame of 35.9% indicating absence of many key interactions required for VEGFR-2 inhibition, hence low pIC₅₀ value.

The flexibility of the protein structure was also studied by plotting the RMSF graph against the protein residues in all the three complexes (Fig. 9). The overall pattern of the fluctuations was observed to be similar in all the cases with little variation. More intense and greater number of fluctuations were observed in sorafenib-receptor complex between 140–260 residues while in case of DDSL6j and SM33 it was similar. The overall pattern of the fluctuations suggests that the protein residues remain rigid throughout the simulation period with very less deviation from their

Table 4 ADMET profile of designed compounds

Comp ID	Absorption		Distribution		METABOLISM						Breast cancer resistance protein
	Human oral bioavailability	Human intestinal absorption	BBB permeability	CNS permeability	Substrate (yes/no)			Inhibitor (yes/no)			
					CYP 1A2	CYP 2C19	CYP 2C9	CYP 1A2	CYP 2C19	CYP 2C9	
DDSL6a	Yes	Yes	No	- 2.96	No	No	No	Yes	No	Yes	No
DDSL6b	Yes	Yes	No	- 3.04	No	No	No	Yes	No	Yes	No
DDSL6c	Yes	Yes	No	- 3.02	No	No	No	Yes	No	Yes	No
DDSL6d	Yes	Yes	No	- 3.05	No	No	No	Yes	No	Yes	No
DDSL6e	Yes	Yes	No	- 2.70	No	No	No	Yes	Yes	Yes	No
DDSL6f	Yes	Yes	No	- 2.68	No	No	No	Yes	Yes	Yes	No
DDSL6g	Yes	Yes	No	- 2.66	No	No	No	Yes	Yes	Yes	No
DDSL6h	Yes	Yes	No	- 2.59	No	No	No	Yes	Yes	Yes	No
DDSL6i	Yes	Yes	No	- 2.61	No	No	No	Yes	Yes	Yes	No
DDSL6j	Yes	Yes	No	- 2.64	No	No	No	Yes	Yes	Yes	No
DDSL6k	Yes	Yes	No	- 2.61	No	No	No	Yes	Yes	Yes	No
DDSL6l	Yes	Yes	No	- 2.56	No	No	No	Yes	Yes	Yes	No
Sorafenib	Yes	Yes	Yes	- 2.10	No	No	Yes	No	Yes	Yes	Yes
Comp ID	EXCRETION			TOXICITY							
	Clearance (ml/min/kg)		Organic cation transporter 2	Half-life of drug (hours)		AMES mutagenesis		Carcinogenesis	Eye corrosion		hERG blockers
DDSL6a	3.89		No	≥ 3	No	No	No	No	No	No	No
DDSL6b	4.00		No	≥ 3	No	No	No	No	No	No	No
DDSL6c	4.17		No	≥ 3	No	No	No	No	No	No	No
DDSL6d	3.70		No	< 3	No	No	No	No	No	No	No
DDSL6e	4.54		No	< 3	No	No	No	No	No	No	No
DDSL6f	4.57		No	< 3	No	No	No	No	No	No	No
DDSL6g	4.59		No	< 3	No	No	No	No	No	No	No
DDSL6h	4.15		No	< 3	No	No	No	No	No	No	No
DDSL6i	4.27		No	< 3	No	No	No	No	No	No	No
DDSL6j	4.37		No	< 3	No	No	No	No	No	No	No
DDSL6k	4.39		No	< 3	No	No	No	No	No	No	No
DDSL6l	3.89		No	< 3	No	No	No	No	No	No	No
Sorafenib	7.02		No	< 3	No	No	No	No	No	No	Yes

initial position. Additionally, Dynamic cross-correlation matrix (DCCM) of residual backbone C α -atoms was used to explore internal collective motion of the protein residues. The DCCM plots are displayed in Fig. 10, where red region indicates correlated motions, blue indicates anti-correlated motions, and white regions indicates no-correlation.

These findings from the RMSD, RMSF and cross correlation matrix plots indicate that the ligand–protein complex between DDSL6j and VEGFR-2 is a stable complex with consistent hydrogen bonding, strong aromatic interactions, and significant hydrophobic contacts.

3.6 MM/GBSA

The results of MM/GBSA as depicted in Table 3 suggests that electrostatic interactions and polar solvation energy are the major contributors of the free energy change while Van der waals interaction had a very little part in it. Other energy components had zero contribution and does not significantly affect the total free energy of the DDSL6j-receptor complex.

3.7 ADMET profiling

The results of the ADMET profiling of the designed compounds, DDSL6(a-l) are enlisted in Table 4. All the compounds were found to possess good oral bioavailability and intestinal absorption which suggests these compounds can be administered orally. In addition to this, none of the compounds showed BBB permeability. This property indicates no CNS side effects. Low log PS value indicates less CNS permeability hence none of the compounds possess CNS toxicity. CYP1A2, CYP2C19 and CYP2C9 are the key enzymes responsible for drug metabolism. None of the compounds act as substrate for these enzymes which indicates no drug-drug interactions whereas sorafenib is a substrate for CYP2C9 and may possess some drug interaction. The breast cancer resistance protein is not inhibited by any of the designed compounds. In terms of excretion, all compounds showed moderate clearance rate (3.70–4.59 ml/min/kg). Moreover, compounds DDSL6a-c showed half-life of more than 3 h while the rest showed less than 3 h. Organic cation transporter 2 was not found to be inhibited by the designed compounds that suggests no drug-drug interaction. Concluding with the toxicity prediction, none of the compounds possess any kind of adverse toxicity like AMES, carcinogenesis, eye corrosion and hERG blockers. Hence all the compounds possess good ADME profile with no toxicity.

4 Conclusion

VEGFR-2 regulates angiogenesis that plays a pivotal role in development of metastatic cancer. Hence, VEGFR-2 is considered as a major target in the treatment of cancer. Quinazoline is a versatile nucleus that has a great potential in development of molecules capable of inhibiting VEGFR-2. Therefore, the principle objective of this study was to construct reliable QSAR models by utilizing the Monte Carlo algorithm of CORALSEA software capable to predict the VEGFR-2 inhibition activity. Out of the six models developed, model M4 was identified as the best fit model based on the statistical values of R^2 , MAE, R_m^2 . The structural attributes including promoters and hinderers extracted from the best fit model were utilized to modify the least potent compound from the dataset SM33 for designing a new series of compounds DDSL6(a-l). Out of these, compound DDSL6j is identified as the most potent VEGFR-2 inhibitor. Amongst all designed molecules, DDSL6j exhibited binding affinity better than sorafenib during molecular docking studies. Finally, MD simulation revealed that DDSL6j forms a more stable complex than both sorafenib and SM33. The ADMET profiling of the designed compounds was also done. All the compounds possessed good pharmacokinetics profile. These findings suggest that quinazoline nucleus can be further explored and optimized to design novel VEGFR-2 inhibitors as potential antiangiogenic agents.

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Author contributions Mayank Kashyap & Saurabh Gupta: conceptualization, data curation, software, methodology, formal analysis, writing—original draft, visualization, validation. Yogita Bansal: project administration, Supervision, review & editing, Gulshan Bansal: validation, formal analysis, and editing.

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Declarations

Ethics approval and consent to participate As this study did not involve human subjects or animals, ethics approval and consent to participate were not required.

Consent for publication All authors have given their consent for the publication of this manuscript. No additional consent for publication was required from individuals.

Competing interests The authors declare no competing interests.

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